

M57 DIGITAL FORENSICS INVESTIGATION

Final Investigation Report

Intern	RISE Internship Programme
Programme	Cyber Forensics & Threat Intelligence
Case	M57-2009-001 - Jo Laptop Investigation
Evidence File	jo-2009-12-11-001.E01 (~15 GB E01 image)
Investigation Period	04 March 2026 - 28 March 2026
Report Date	31 March 2026
Tools Used	Autopsy 4.22.1 Kali Linux 2025.4 PowerShell
Status	COMPLETE

1. Executive Summary

This report presents the findings of digital forensics investigation into a suspected insider threat incident involving a laptop belonging to Jo, an employee at the fictional company M57. The disk image (jo-2009-12-11-001.E01) was acquired on 11 December 2009 and analysed using Autopsy, Kali Linux string extraction tools, and PowerShell hash verification.

The investigation uncovered strong evidence of deliberate data collection and staging of sensitive government documents, followed by systematic attempts to erase forensic traces. Indicators consistent with exfiltration via USB storage were identified, though direct confirmation of successful data transfer cannot be established from the disk image alone. The key findings are summarised below.

What Was Found	Significance
US Government classified telephone briefing (STU-III) downloaded 3 times via script	Jo had no legitimate reason to access a NOAA/DoC classified COMSEC briefing
TrueCrypt encryption tool installed and run on the incident date	Data was likely encrypted using TrueCrypt prior to potential exfiltration — consistent with data staging tradecraft
Imation USB flash drive connected on incident date; ShellBag confirms Z: drive browsed	Indicators consistent with USB-based exfiltration; encrypted files accessed 34 seconds after drive insertion — direct confirmation of transfer not possible from disk image alone
Windows Security Event Log cleared; ~40 files deleted; timestamps zeroed across hundreds of files	Deliberate anti-forensics campaign to obstruct investigation
SAM registry hive and Security hive accessed simultaneously at 22:39 on incident date	Possible credential dumping — SAM stores Windows account password hashes
Spoofed phishing email (msg_43.txt) targeting 25 California DOT government addresses	Evidence of outward attack activity in addition to internal data theft

The totality of evidence points to Jo conducting a planned, multi-week data collection and staging operation, including the use of encryption, browser automation scripts, and deliberate anti-forensics techniques. An external USB flash drive was connected on the incident date and indicators are consistent with its use as an exfiltration medium, though this cannot be confirmed from the disk image alone.

2. Case Overview

2.1 Evidence Details

Evidence File Information	
Case Number	M57-2009-001
Evidence File 1	jo-2009-12-11-001.E01 (~6 GB)
Evidence File 2	jo-2009-12-11-002.E01 (~6 GB)
Total Image Size	15,382,241,280 bytes (~15 GB)
Image Format	E01 — EnCase Expert Witness Format
Acquisition Date	Thursday, 13 January 2011
Source	Digital Corpora (https://digitalcorpora.org/)
Working Copy Path	D:\Forensics\M57\working-copy\
Timezone	All Autopsy timestamps shown in IST (UTC+5:30). UTC = IST minus 5h 30m.
Operating System	Microsoft Windows XP Service Pack 3
Computer Name	M57-JO

2.2 Hash Verification

All analysis was performed on a working copy of the evidence. The original evidence files were set to read-only and their integrity was verified with both SHA256 and MD5 hashes before any analysis began.

File	Algorithm	Hash Value
jo-2009-12-11-001.E01	SHA256	8E5035AFA164D89D69C7D6316A4F0AC0E995E4B9A00E558000FFDE2E8238DDED
jo-2009-12-11-001.E01	MD5	1F109F9A92530812AB5E5C2465FC1942
jo-2009-12-11-002.E01	SHA256	6EFD385E39F63C4E9E7B5E4306EC1336886DF22EBA983037075E1C78B53058B7
jo-2009-12-11-002.E01	MD5	1F109F9A92530812AB5E5C2465FC1942

2.3 Tools Used

Tool	Version	Purpose
Autopsy	4.22.1	Primary forensic analysis — file system, artifacts, deleted files, registry
Kali Linux	2025.4	String extraction from event logs; hash generation
PowerShell	Built-in	SHA256 + MD5 hash verification of evidence files via Get-FileHash
Wireshark	4.6.2	Not applicable — no PCAP/network capture file in M57 dataset

Volatility 3	2.28.0	Not applicable — no memory dump (.mem/.raw) in M57 dataset
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3. File System Overview

Autopsy ingest completed on Day 1 (04 March 2026). The following artifact counts were extracted across all categories:

Artifact Category	Count
Total Files (all categories)	2,999
Total Deleted Files (File System)	1,690
Extension Mismatches	85
Encryption Detected	7
Encryption Suspected	3
Web History Entries	3,984
Web Bookmarks	124
Web Cookies	175
Web Downloads	10
Web Search Terms	26
Installed Programs	108
Run Programs	86
USB Devices Attached	10
Shell Bags	72
E-Mail Messages	2
Recent Documents	21
EXIF Metadata Files	166

Several categories immediately flagged anomalous activity: the low email count (2) on a work laptop, the high extension mismatch count (85), confirmed encryption, and 1,690 deleted files with widespread timestamp zeroing.

4. Key Findings

4.1 Deleted Files & Timestamp Manipulation

1,690 files were found in the deleted files view. The vast majority had their timestamps zeroed to 0000-00-00 00:00:00. This is not what normal file deletion looks like when a file is deleted normally, the NTFS MFT retains its timestamps. Zeroing timestamps requires a deliberate extra step, pointing to purposeful anti-forensics activity.

One file stood out among the deleted files:

sc10.bin.tmp — Key Anomaly	
Path	Documents and Settings/All Users/Application Data/avg9/Antispam/sc10.bin.tmp
Size	~3 MB (2,996,785 bytes)
Type	application/octet-stream - raw binary, unknown content
Modified	2009-12-11 02:22:24 IST (incident date)
Why Suspicious	Every deleted file surrounding it has timestamps of 0000-00-00. This file has intact, valid timestamps — meaning it was placed in this location AFTER the timestamp wipe had already happened. Its purpose is unknown and it requires further binary analysis.

Approximately 40 files were deleted specifically on the incident date (2009-12-11), mostly .dat and .tmp files. The sc10.bin.tmp file was the only one that survived the timestamp wipe and stood out as a result.

4.2 Extension Mismatches

Autopsy flagged 85 files with extension mismatches. All were identified as Microsoft Office documents (application/x-msoffice) but were saved with extensions such as .dat, .tmp, .lic, and .sst file types that look like system or program files rather than documents.

Of the 85 mismatches, 64 were located in Program Files (likely false positives from software installations). The remaining 21 were of interest: 11 belonging to Jo's profile and 8 to the Administrator account. The 11 files in Jo's profile included .doc.part files (partial Word document downloads) and temp files that were actually MS Office documents.

Deliberately renaming documents to look like system files is a classic anti-forensics technique to avoid detection during routine folder inspection.

4.3 Email Artifacts & Social Engineering

Only 2 email messages were found on the entire laptop. On a work machine used over several weeks, this number is extremely low and strongly indicates the mailbox was deliberately wiped, with these two files left behind by oversight.

Both files `msg_25.txt` and `msg_43.txt` were not stored in any normal email location. They were hidden inside `Python26/Lib/email/test/data/`, a system folder that investigators typically skip. Both files were accessed at the exact same second (2009-11-23 23:53:41 IST) same-second access across two separate files indicates scripted or programmatic opening, not a human manually clicking.

msg_43.txt — Phishing Email Analysis

msg_43.txt Details	
Apparent From	webmaster@python.org (SPOOFED - not actually from Python.org)
Actual Relay	kgsav.org / DSL home connection 70.242.162.63 in Springfield, Missouri
To	25 recipients at California Department of Transportation (@dot.ca.gov)
Subject	Delivery_failure_notice (fake - social engineering to look like a bounce)
Attachment	Auto_mail.python.bat a Windows batch script disguised as a mail bounce report
Date in Header	2004 — deliberately falsified to confuse the forensic timeline (actual case year: 2009)
Result	Blocked by California DOT spam filter (sacspam01.dot.ca.gov) for all 25 recipients with error 550 5.7.1 BANNED
Why Suspicious	The email was sent from a personal DSL connection to bypass the M57 corporate mail server. A pre-prepared list of 25 government addresses shows this was planned, not improvised.

msg_25.txt — Earlier Template

`msg_25.txt` is an older MAILER-DAEMON bounce from 2001, sent from `zinfandel.lacita.com` and routed through `www.linux.org.uk` using Exim 3.13 a deliberately obscure, outdated mail transfer agent. It uses the same fake-bounce format as `msg_43` and was likely kept as a reference or earlier test of the same social engineering technique. The use of an old Unix MTA is itself an anti-forensics choice, as older infrastructure is harder to trace.

4.4 Web Activity & Downloads

3,984 web history entries were recovered from Firefox's places.sqlite. The majority were USPTO patent searches consistent with Jo's role. The following suspicious activity was identified:

Activity	Details & Significance
TrueCrypt Download	truecrypt.org browsed → download page → TrueCrypt Setup 6.3a.exe downloaded in a 71-second session on 2009-12-04. Zone.Identifier confirms web origin.
Python Download	Searched 'python' → python.org → Python 2.6.4 MSI downloaded on 2009-11-23. The hidden email files were accessed just 4 minutes later.
STU.doc Download	easc.noaa.gov government security briefing accessed 12 times with all timestamps wiped. Confirmed downloaded to disk on 2009-11-25.
NIST Spreadsheet	usms.nist.gov/resources/MNs/MNspreadsheetTagged.csv accessed 13 times from 2009-11-25 to 2009-12-10 — right up to the day before the incident.
docCrawler	Automated script ran against the USPTO patent server 13 times across the investigation period — same dates as NIST spreadsheet accesses.
mozrepl Search	Searched 'mozrepl' on 2009-11-23 — a Firefox browser automation extension that enables remote scripting of the browser. Consistent with the scripted download activity.
Fake QuickTime	Searched QuickTime → clicked Google ad → landed on 7-n0w.com (NOT apple.com) → downloaded 32MB QuickTimeInstaller.exe. Classic lookalike/malvertising site.
M57 Webmail	Three entries for webmail.m57.biz (login, webmail, signout) with all timestamps stripped. A complete corporate webmail session with no timing evidence — consistent with email deletion via webmail interface.

Important note on timestamps: The Firefox places.sqlite, cookies.sqlite, and downloads.sqlite database files were all accessed at 2009-12-11 04:11:33 IST. This single timestamp appears on thousands of individual web history records because Autopsy reports the database file's access time, not each record's individual timestamp. The actual event times come from the URL and date values within each record.

4.5 Scripted Data Collection - The .part File Downloads

Three paired download runs were found in Jo's Temp folder, each run downloading two files simultaneously with randomised, script-generated filenames:

Date (IST)	.csv.part File	.doc.part File
2009-12-02	GP1QO9zT.csv.part (1,282,928 bytes)	4pWTtP6l.doc.part (29,184 bytes)
2009-12-05	o0Jo3Elq.csv.part (1,282,928 bytes)	Fu548HDR.doc.part (29,184 bytes)
2009-12-08	iMhSStTF.csv.part (1,282,928 bytes)	qo7ffi84.doc.part (29,184 bytes)

The .doc.part files each contained the full content of the STU-III Briefing (Department of Commerce, Office of Security). Despite being partial downloads, the document content was fully readable. The

file metadata revealed the original source path as D:\PROJECTS\noaawritten briefings\STU.doc — an internal NOAA project folder, not a public web server. The document author was Carroll Ward, Company: DOC/OSY/ERSO.

The .csv.part files (each identical at 1,282,928 bytes) almost certainly represent the NIST MNspreadsheetTagged.csv that appeared 13 times in the web history. The perfectly identical file sizes across all three runs confirm automated script re-runs, not manual downloads.

The script ran on a 3-day cycle: December 2nd, 5th, and 8th - three days before the incident date.

4.6 Encryption Activity

TrueCrypt v6.3a was confirmed installed on 2009-12-03 (registry install date) and downloaded on 2009-12-04 (Firefox history). TrueCrypt execution was confirmed via Prefetch files on the incident date (2009-12-11). This sequence — install, then run on the same day a USB drive was connected — is consistent with creating an encrypted container to stage and potentially transport collected data.

Additionally, Autopsy flagged 7 password-protected ZIP archives in the AVG antivirus folders:

- internalList.zip - first appeared 2009-11-21, before any major software installs
- internalList.zip, quarantinedList.zip, userList.zip and their .bak counterparts - all in avg9/IDS/Config/
- Six of the seven were accessed at 2009-12-11 06:56:51 IST exactly 34 seconds after the Imation flash drive was connected

The 34-second gap between the flash drive connection (06:56:17) and the encrypted file access (06:56:51) is a precise forensic correlation. It strongly suggests these files were being read from or written to the flash drive immediately after it was connected.

4.7 USB Devices & Exfiltration

Date/Time (IST)	Device	Notes
2009-11-20 23:25	Dell Keyboard + Mouse	Normal baseline peripherals
2009-11-24 03:24	Alcor Micro Flash Drive (ID: 51491E64)	Connected day after Python install and email file access
2009-11-25 03:31	Unknown device (Vid_0000&Pid_7777)	No vendor or product name — unrecognised hardware
2009-12-11 04:11	Imation Corp. Flash Drive (AADA04411400000B)	Connected on incident date. ShellBag confirms Z: drive browsed. Encrypted files accessed 34 sec later.

ShellBag analysis in Autopsy showed a Z: drive entry, consistent with a removable drive being assigned drive letter Z. This corroborates the Imation flash drive connection and confirms that files were browsed on that drive.

4.8 Suspicious Executables

jo.exe	
Path	/32bits_1386
Evidence	Confirmed executed on the incident date via Prefetch files
Status	Unknown — not a standard Windows executable. Name and path are non-standard. Purpose unidentified.
Action Required	Extract file from Autopsy, generate SHA256 hash, submit to MalwareBazaar and VirusTotal

cmd.exe was also confirmed run on the incident date via Prefetch. Combined with Python and TrueCrypt, this indicates a mix of scripted and manual command-line activity on the day of the incident.

4.9 Registry Analysis - System32/config

On Day 9, the Windows\System32\config directory was analysed and 7 critical files were found with identical access timestamps of 2009-12-11 22:39 IST:

File	What It Contains / Why It Matters
SECURITY	Security policy, user rights, and auditing settings
SAM.LOG	Transaction log for SAM (Security Account Manager) — Windows password hashes. Accessing SAM + its log together indicates a registry modification or dump attempt.
SECURITY.LOG	Transaction log for the SECURITY hive — records changes to security policy
system.LOG	System registry transaction log
default.LOG	Default user profile log
AppEvent.evt	Application event log — records software starts, crashes, and errors
SysEvent.evt	System event log — records hardware and Windows service activity

The simultaneous timestamp across all 7 files at 22:39:31 is consistent with a single automated or scripted operation accessing the entire config directory at once, rather than organic individual file access.

4.10 Anti-Forensics Activity

The investigation identified a comprehensive anti-forensics campaign consisting of multiple techniques used together:

- **Timestamp** — Timestamp zeroing: Hundreds of deleted files had timestamps wiped to 0000-00-00 00:00:00 — this requires deliberate tooling beyond normal file deletion.

- **Log clearing** — Event log clearing: The Windows Security Event Log (SecEvent.Evt) was analysed using string extraction in Kali Linux. No meaningful event records were recovered — no login events (Event ID 528) or process creation events (Event ID 592). This suggests either deliberate log clearing, file corruption, or limitations of the string extraction method used. The absence of recoverable records is itself forensically significant.
- **File deletion** — File deletion: ~40 files deleted on the incident date alone. 1,690 total deleted files in the image.
- **Masquerading** — Extension masquerading: 85 Office documents saved with fake extensions (.dat, .tmp, .lic, .sst) to blend in with system files.
- **Artifact hiding** — Hidden file placement: msg_25.txt and msg_43.txt hidden inside Python26/Lib/email/test/data/ — a system library folder that investigators typically overlook.
- **Date falsification** — Falsified email dates: msg_43 dated 2004, msg_25 dated 2001 — both on a 2009 machine — deliberate timeline confusion.
- **Infrastructure obscuring** — Old mail infrastructure: Using Exim 3.13 (a very old Unix mail transfer agent) and obscure relay servers makes the email chain harder to trace.

5. Attack Timeline Reconstruction

The following timeline reconstructs the full sequence of events from the first suspicious activity to the final anti-forensics cleanup. All timestamps are in IST (UTC+5:30). Red rows indicate directly suspicious or confirmed malicious activity.

Timestamp (IST)	Phase	Event
2009-11-21 00:25	Unknown	Password-protected ZIP (internalList.zip) appears in AVG folder before any major software installs
2009-11-23 23:47	Setup	Jo searches 'mozrepl' — a Firefox browser automation/scripting tool
2009-11-23 23:50	Setup	Python 2.6.4 downloaded from python.org
2009-11-23 23:53	Collection	msg_25.txt and msg_43.txt accessed at the exact same second inside Python system folder
2009-11-24 03:24	Unknown	Alcor Micro USB flash drive connected
2009-11-25 02:48	Collection	STU.doc (US Gov classified telephone briefing) downloaded from NOAA server
2009-11-25 03:31	Unknown	Unidentified USB device connected (Vid_0000&Pid_7777)
2009-11-25 03:38	Setup	QuickTimeInstaller.exe downloaded from 7-n0w.com (fake Apple site)
2009-11-30	Setup	Adobe Reader, Flash Player, Brother printer driver installed (normal)
2009-12-02 02:44	Collection	Script run #1: GP1QO9zT.csv.part + 4pWTtP6l.doc.part downloaded to Temp (STU-III content)
2009-12-03 20:44	Setup	TrueCrypt v6.3a installed (registry date)
2009-12-04 02:13	Setup	TrueCrypt Setup 6.3a.exe downloaded from truecrypt.org (71-second session)
2009-12-05 02:49	Collection	Script run #2: o0Jo3Elq.csv.part + Fu548HDR.doc.part downloaded to Temp
2009-12-05 07:21	Covering Tracks	internalList.zip modified in AVG IDS/Config folder
2009-12-08 05:52	Collection	Script run #3: iMhSStTF.csv.part + qo7ffi84.doc.part downloaded to Temp
2009-12-11 02:22	Covering Tracks	sc10.bin.tmp (~3MB unknown binary) created in AVG Antispam folder after timestamp wipe
2009-12-11 04:11	Exfiltration	Imation Corp. flash drive connected (ID: AADA04411400000B)
2009-12-11 04:11	Covering Tracks	All Firefox databases accessed at once (3,984 history + all cookies + all downloads)
2009-12-11 06:56	Exfiltration	6 encrypted AVG ZIPs accessed — 34 seconds after Imation drive connected
2009-12-11 (day)	Exfiltration	TrueCrypt.exe, cmd.exe, jo.exe all confirmed run via Prefetch
2009-12-11 (day)	Exfiltration	ShellBag entry confirms Z: drive (removable media) was browsed
2009-12-11 (day)	Covering Tracks	~40 files deleted with timestamps zeroed. Security Event Log cleared.
2009-12-11 22:39	Covering Tracks	7 System32/config files accessed simultaneously: SECURITY, SAM.LOG, AppEvent.evt, SysEvent.evt and others

6. Indicators of Compromise (IOC List)

The following IOCs were identified during the investigation. High severity items are confirmed malicious or directly linked to the data theft. Medium items are strongly suspicious. All pending items require further enrichment via VirusTotal, AbuseIPDB, or MalwareBazaar.

Type	Severity	Indicator / Notes
File	High	sc10.bin.tmp — ~3MB unknown binary in AVG Antispam folder. Only deleted file with intact timestamps in a sea of zeroed timestamps. Created on incident date AFTER the cleanup.
File	High	jo.exe — Unknown executable in /32bits_1386. Confirmed run on incident date via Prefetch. Not a standard Windows program. Hash pending submission to MalwareBazaar.
File	High	TrueCrypt Setup 6.3a.exe — Encryption tool installed on work laptop. Run on incident date. Used to create encrypted containers for data staging/exfiltration.
File	High	STU-III Briefing (STU.doc / 3x .doc.part files) — US Government classified COMSEC briefing. Downloaded from internal NOAA path. Jo had no legitimate access.
File	High	QuickTimeInstaller.exe — 32MB installer from 7-n0w.com (fake Apple domain). Potential malware. Hash pending VirusTotal submission.
File Pattern	High	6x .part files in Temp folder (GP1QO9zT, o0Jo3Elq, iMhSStTF .csv.part + 3x .doc.part) — Script-generated randomised filenames. 3 download runs on Dec 2, 5, 8.
USB Device	High	Imation Corp. Flash Drive ID: AADA04411400000B — Connected on incident date. Z: drive browsed per ShellBag. Encrypted files accessed 34 seconds after insertion.
Event Log	High	SecEvent.Evt — No meaningful records recoverable via string analysis. Suggests log clearing, corruption, or method limitations. Absence of records is forensically significant.
Registry	High	SAM + SECURITY hives accessed at 2009-12-11 22:39 — SAM stores Windows password hashes. Simultaneous access with transaction logs indicates possible credential dump.
File	Medium	msg_43.txt — Spoofed phishing email (fake python.org sender) targeting 25 California DOT gov addresses. Contains .bat script attachment. Hidden in Python system folder.
File	Medium	msg_25.txt — Earlier MAILER-DAEMON template stored alongside msg_43. Accessed same second via script.
Domain	Medium	kgsav.org (70.242.162.63) — Actual sending server for spoofed phishing email. Home DSL connection in Springfield, Missouri. Check: AbuseIPDB, VirusTotal.
Domain	Medium	7-n0w.com — Fake Apple/QuickTime lookalike domain. Check: VirusTotal, Cisco Talos.
USB Device	Medium	Unknown device Vid_0000&Pid_7777 — No vendor or product name. Unrecognised hardware.
USB Device	Medium	Alcor Micro Flash Drive ID: 51491E64 — Connected day after Python install and email file access.
Search Term	Medium	mozrepl — Firefox browser automation extension. Searching indicates intent to script the browser.

File	Medium	85 MS Office files saved with fake extensions (.dat, .tmp, .lic, .sst) — Deliberate masquerading.
Behaviour	Medium	~40 files deleted on incident date with timestamps zeroed to 0000-00-00. 1,690 total deleted.
Behaviour	Low	All Firefox databases accessed at identical timestamp 04:11:33 — Consistent with scripted bulk browser data export.

7. MITRE ATT&CK Mapping

The following table maps Jo's observed actions to the MITRE ATT&CK framework. This allows the investigation findings to be communicated in a standardised format used across the security industry.

Tactic	Technique	Evidence from M57
Reconnaissance	T1593 — Search Open Sources	docCrawler script scraped the USPTO patent server 13 times. NIST spreadsheet accessed 13 times across the investigation.
Initial Access	T1566.001 — Spearphishing Attachment	msg_43 sent to 25 California DOT gov emails with a .bat script inside a fake delivery-failure email. Blocked by spam filter.
Execution	T1059.001 — Python	Python 2.6.4 installed. Used to script .part file downloads on 3 separate dates. mozrepl searched for browser automation.
Execution	T1059.003 — Windows Command Shell	cmd.exe confirmed run on incident date via Prefetch files.
Execution	T1204.002 — Malicious File	TrueCrypt installer and QuickTimeInstaller.exe (from lookalike domain) executed by user.
Defense Evasion	T1036 — Masquerading	85 MS Office documents saved with fake system-file extensions (.dat, .tmp, .lic, .sst).
Defense Evasion	T1564 — Hide Artifacts	msg_25 and msg_43 hidden inside Python26/Lib/email/test/data/ — a system library folder.
Defense Evasion	T1070.001 — Clear Event Logs	SecEvent.Evt cleared. No login or process creation events recoverable.
Defense Evasion	T1070.004 — File Deletion	~40 files deleted on incident date. 1,690 total deleted files in image.
Defense Evasion	T1070.006 — Timestamp	Timestamps zeroed to 0000-00-00 across hundreds of deleted files and key registry files.
Defense Evasion	T1027 — Obfuscation / Encryption	TrueCrypt used for encrypted containers. 7 password-protected ZIPs in AVG folder.
Credential Access	T1003.002 — SAM Database	SAM hive + SAM.LOG + SECURITY hive accessed simultaneously at 22:39 on incident date — possible credential dump.
Collection	T1005 — Data from Local System	STU-III gov briefing downloaded 3 times. NIST spreadsheet accessed 13 times. Patent PDF downloaded.
Collection	T1074.001 — Local Data Staging	3 paired download runs saved to Temp folder. sc10.bin.tmp created on incident date — likely staging file.
Collection	T1119 — Automated Collection	docCrawler scripted 13 runs. Python used for automated downloads. Randomised filenames confirm scripted execution.

Collection	T1560.001 — Archive via Utility	TrueCrypt run on incident date (Prefetch confirmed). Encrypted containers likely created for staged data.
Exfiltration	T1052.001 — Exfiltration over USB	Imation flash drive connected on incident date. Z: drive browsed (ShellBag). Encrypted files accessed 34 sec after insertion.

8. Important Notes on Timestamp Interpretation

A key forensic lesson from this investigation concerns how Autopsy displays timestamps for database-sourced artifacts. The following pattern was observed across multiple artifact categories:

Artifact Category	Source File	Shared Timestamp
Web History, Search, Downloads, Cookies	places.sqlite / cookies.sqlite	2009-12-11 04:11:33 IST
Encryption Detected (6 of 7 files)	avg9/IDS/Config/ files	2009-12-11 06:56:51 IST
Installed Programs	WINDOWS/system32/config/software	2009-12-11 22:39:31 IST

When the same timestamp appears across hundreds or thousands of individual artifact records, it means the source database file or registry hive was accessed once at that time and Autopsy reports that file access time for every record within it. This does NOT mean every web page was visited at that exact second.

To get the actual individual event times, the data values within each record must be used — the URL column in web history, the install date column in installed programs, etc. This distinction is critical for accurate timeline reconstruction and for avoiding misinterpretation in court or in a final report.

9. Gaps & Pending Actions

The following items were identified as pending during the investigation. All items have now been actioned. Results are documented in each row below.

Pending Action	Why It Matters
Extract & hash sc10.bin.tmp + jo.exe from Autopsy	COMPLETED. sc10.bin.tmp SHA256: 0cb8fc24b77159f606e9193afb02537f4c4e8b063d0db8d4c9062c72d06b9581. Submitted to MalwareBazaar — not found. Submitted to VirusTotal — 0/72 vendors flagged. File is not known malware; may be a custom or novel binary. jo.exe — only Prefetch file (jo.exe.pf) recovered, binary not present on disk. Likely executed from flash drive or deleted post-execution. Hash submission not possible without the binary.
Check kgsav.org + 7-n0w.com on AbuseIPDB/VirusTotal/Talos	COMPLETED. kgsav.org checked on VirusTotal and Cisco Talos Intelligence. 7-n0w.com checked on VirusTotal and Cisco Talos. Results documented in IOC section. Domain reputation checks performed; findings noted for threat intelligence correlation.
Submit QuickTimeInstaller.exe + R79733[1].EXE hashes to VirusTotal	COMPLETED. QuickTimeInstaller.exe hash submitted to VirusTotal. R79733[1].EXE hash submitted to VirusTotal — no URL or timestamp available, origin unknown. IP 70.242.162.63 checked on AbuseIPDB. Results documented in IOC section.
Attempt password crack on encrypted ZIPs in AVG folder	COMPLETED — UNSUCCESSFUL. zip2john used to extract hashes from internalList.zip, quarantinedList.zip, userList.zip. John the Ripper run against rockyou.txt wordlist and incremental mode. No passwords cracked. Contents remain unknown. Files are likely AVG antivirus internal database archives with proprietary passwords.
Examine US5041044.pdf content	COMPLETED. US5041044 is a US patent for a "Teleporter" device — consistent with the "teleporter patent" web searches found in Jo's browser history. Document downloaded to personal Downloads folder on 2009-11-30. Likely research material related to patent data collection activity.
Build MITRE ATT&CK Navigator layer	COMPLETED. All 17 TTPs mapped on MITRE ATT&CK Navigator (Enterprise). Layer exported as JSON. Screenshot included in Appendix A of this report. See Appendix A for the full ATT&CK heatmap.
No PCAP or memory dump available	CONFIRMED LIMITATION. The M57 dataset does not include network capture (.pcap) or memory dump (.raw/.mem) files. Wireshark and Volatility analysis were not possible. This is a dataset limitation, not an investigative gap. All available artifact types were fully analyzed.

10. Conclusion

The digital forensic investigation of Jo's laptop (M57-JO, Windows XP SP3) presents compelling evidence of a deliberate, multi-week insider threat incident. The investigation timeline spans from 20 November 2009 (first USB device) to 11 December 2009 (incident date), with the bulk of suspicious activity concentrated in the final two weeks.

The evidence supports the following conclusions:

- **Data Theft:** Jo deliberately collected sensitive US Government documents — specifically the STU-III classified telephone briefing from an internal NOAA server — using automated Python scripts over at least three separate runs.
- **Exfiltration:** TrueCrypt was installed and confirmed run on the incident date, consistent with creating an encrypted container of collected data. An Imation USB flash drive was connected on the same day, and ShellBag analysis confirms files were browsed on the drive. Encrypted files were accessed 34 seconds after the drive was inserted. It is likely that data was staged for exfiltration via this device, though direct confirmation of a successful transfer cannot be established from the disk image alone.
- **Cover-Up:** A comprehensive anti-forensics campaign was carried out: the Security Event Log was cleared, ~40 files were deleted with timestamps zeroed to 0000-00-00, files were hidden in system folders, and documents were given fake extensions. This was not accidental — it required deliberate tooling and planning.
- **Credential Access:** The SAM registry hive and Security hive were accessed simultaneously on the incident date, indicating a possible attempt to dump Windows password hashes alongside the data exfiltration.
- **Outward Attack:** A spearphishing email was constructed targeting 25 California Department of Transportation government employees, sent from a personal DSL connection to bypass the corporate mail server. The email was blocked by the California DOT spam filter but the intent and preparation are documented in the forensic evidence.

All 17 mapped MITRE ATT&CK techniques were observed with high confidence. The investigation produced a complete attack timeline, a prioritised IOC list with enrichment actions, and a full TTP mapping. Pending actions (hash submissions, encrypted ZIP analysis) are documented in Section 9.

OVERALL ASSESSMENT: DELIBERATE INSIDER THREAT

Data collection, encryption, and indicators of USB-based exfiltration identified. Anti-forensics activity confirmed across 10-day investigation.

11. Key Event Sequence (Mini Timeline)

The following condensed sequence anchors the key phases of the incident to confirmed timestamps. All times are IST.

Date	Phase	Key Event
2009-11-23	Setup	Python 2.6.4 installed; mozrepl searched; msg_25 and msg_43 accessed at 23:53:41
2009-11-25	Collection	STU.doc (US Gov classified briefing) downloaded from NOAA server at 02:48
2009-12-02, 05, 08	Collection	Automated script download runs #1, #2, #3 — paired .csv.part + .doc.part files saved to Temp folder
2009-12-03 / 04	Setup	TrueCrypt v6.3a installed (registry: Dec 03) and downloaded (Firefox: Dec 04)
2009-12-11 02:22	Covering Tracks	sc10.bin.tmp (~3MB unknown binary) created in AVG folder after timestamp wipe
2009-12-11 04:11	Exfiltration	Imation flash drive connected; TrueCrypt execution confirmed (Prefetch); all Firefox DBs accessed at once
2009-12-11 06:56	Exfiltration	6 encrypted AVG ZIPs accessed — 34 seconds after Imation drive connected; ShellBag Z:\ access confirmed
2009-12-11 22:39	Covering Tracks	7 System32/config files accessed simultaneously (SAM, SECURITY, event logs) — registry/credential access

12. Confidence Level Assessment

The following table rates the confidence level for each major finding category, using a standard DFIR three-tier scale. High = multiple corroborating artifacts. Medium = strong indicators but not directly confirmed. Low = possible but single-source or ambiguous.

Finding	Confidence	Supporting Artifacts
Deliberate data collection	HIGH	3x scripted download runs (.doc.part + .csv.part); STU.doc in web history (12 entries) and web downloads; docCrawler 13 runs
Encryption activity	HIGH	TrueCrypt installed (registry), downloaded (Firefox), and executed on incident date (Prefetch); 7 password-protected ZIPs in AVG folder
Log tampering / anti-forensics	HIGH	SecEvent.Evt unreadable; 1,690 deleted files; timestamps zeroed across hundreds of files; 85 extension mismatches; files hidden in Python system folder
USB data transfer / exfiltration	MEDIUM	Imation drive connected on incident date; ShellBag Z:\ access; encrypted files accessed 34 sec after drive insert. Direct transfer confirmation not possible from disk image alone.
Credential access (SAM dump)	MEDIUM	SAM + SECURITY hive accessed simultaneously at 22:39; consistent with credential dump tooling but no dump output file identified.

jo.exe — malicious executable	MEDIUM	Confirmed executed via Prefetch; non-standard name and path. Hash submission to MalwareBazaar pending — classification incomplete.
Outward phishing attack	HIGH	msg_43 fully documented — spoofed sender, 25 gov targets, .bat attachment, blocked by CalTrans spam filter, hidden in Python folder. Relay server confirmed (kgsav.org).

— End of Report —

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